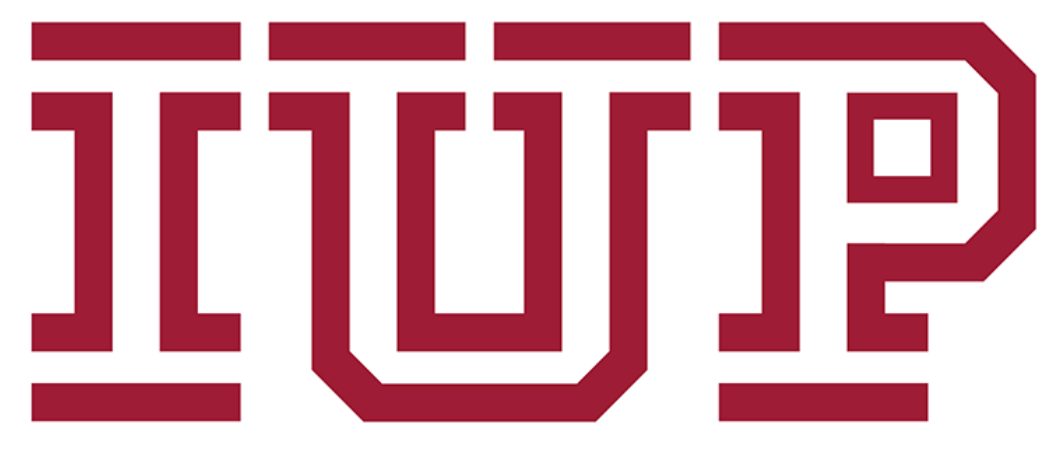




Clustering and Recommending Songs with Machine Learning

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Abstract

The goal of this research project is to leverage a variety of machine-learning techniques to:

- Transform Music Clips to fixed-length vector embedding
- Visualize Music Clips as Points in 2d Space
- Group Music into different categories based on the vector embedding
- Create a recommendation based on a set of liked songs

Potential Applications

This research project has many potential applications. Although most of the applications would require a larger modern dataset. some of the potential applications include:

- Music streaming services
- Automatically Cataloging Song
- Exploring the similarity of 2 songs
- Identifying shifts in music genres over time

Dataset Description

The dataset used in this research project is the GTZAN audio dataset. The dataset contains 1000 tracks of 30-second samples of various music genres (100 each). All the tracks are in a consistent format sample rate of 22050Hz, mono-channel 16-bit audio files saved in the .wav file format

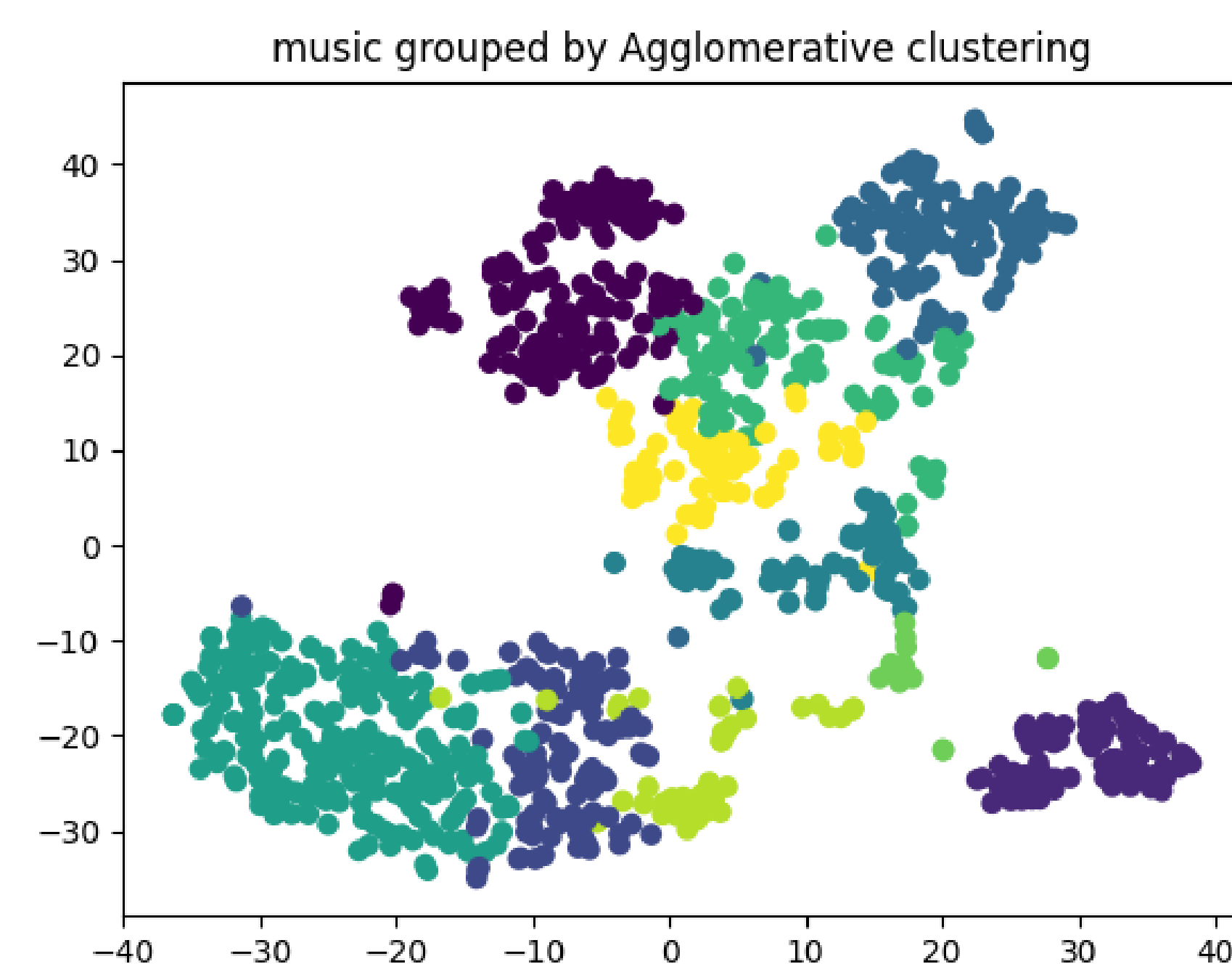
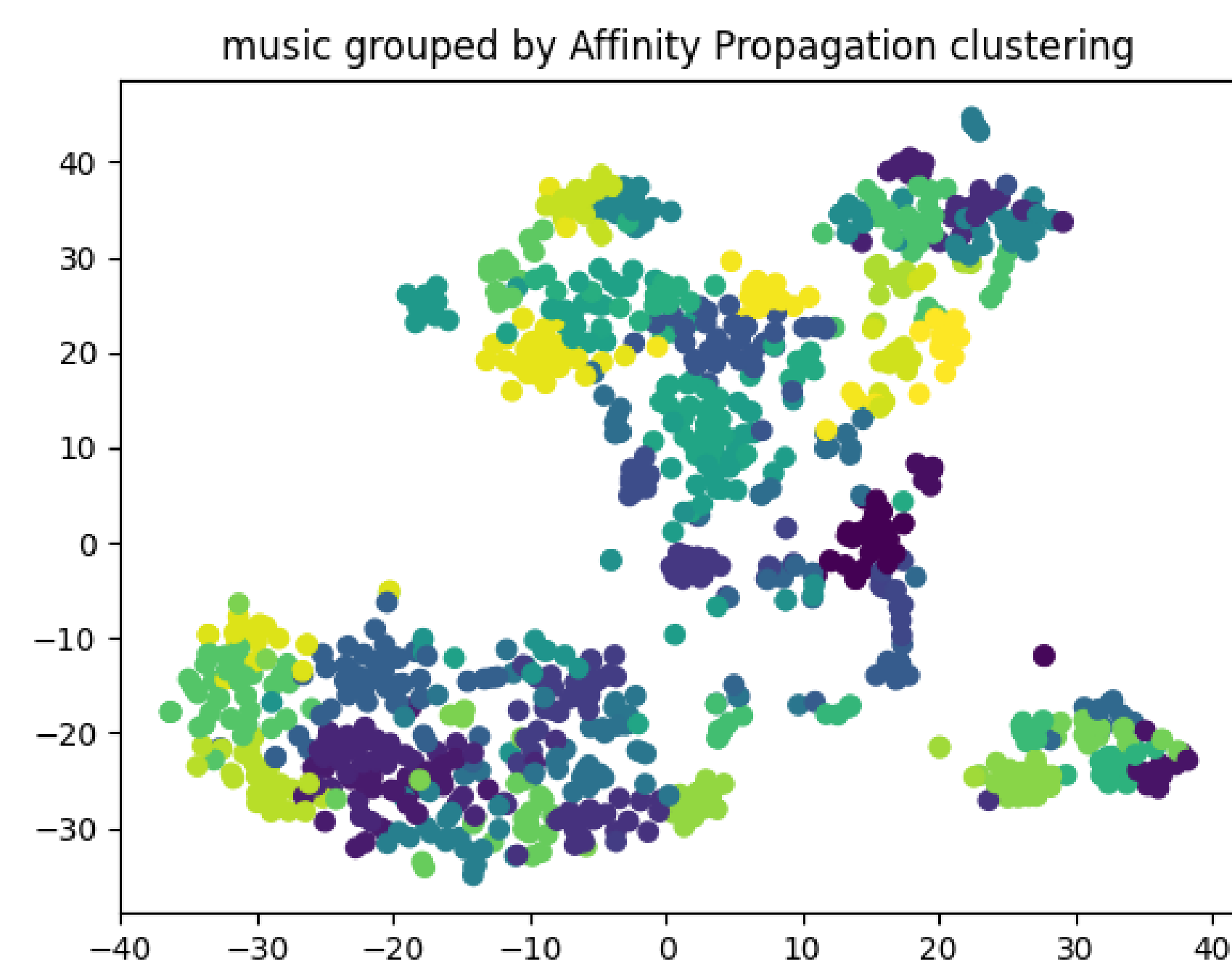
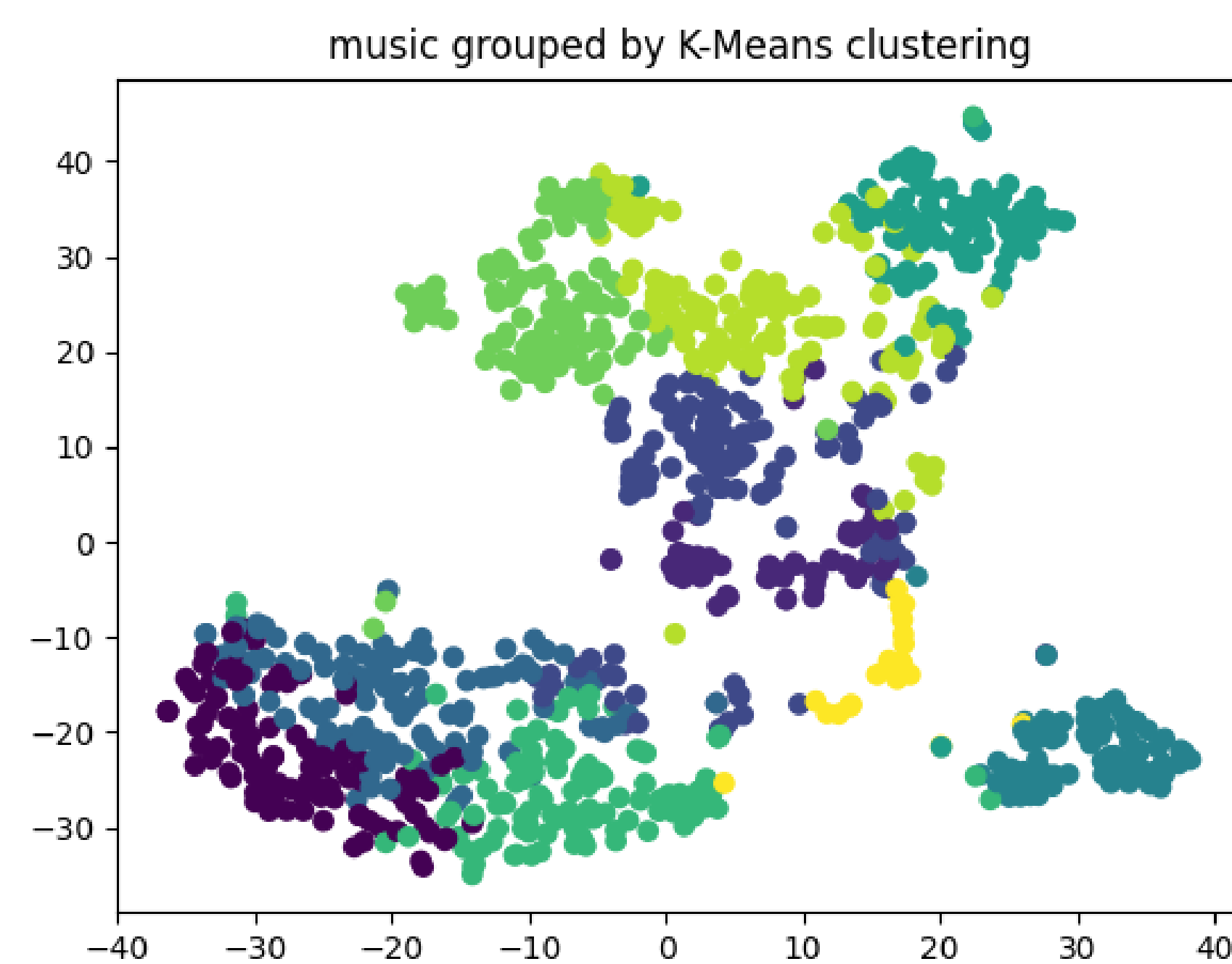
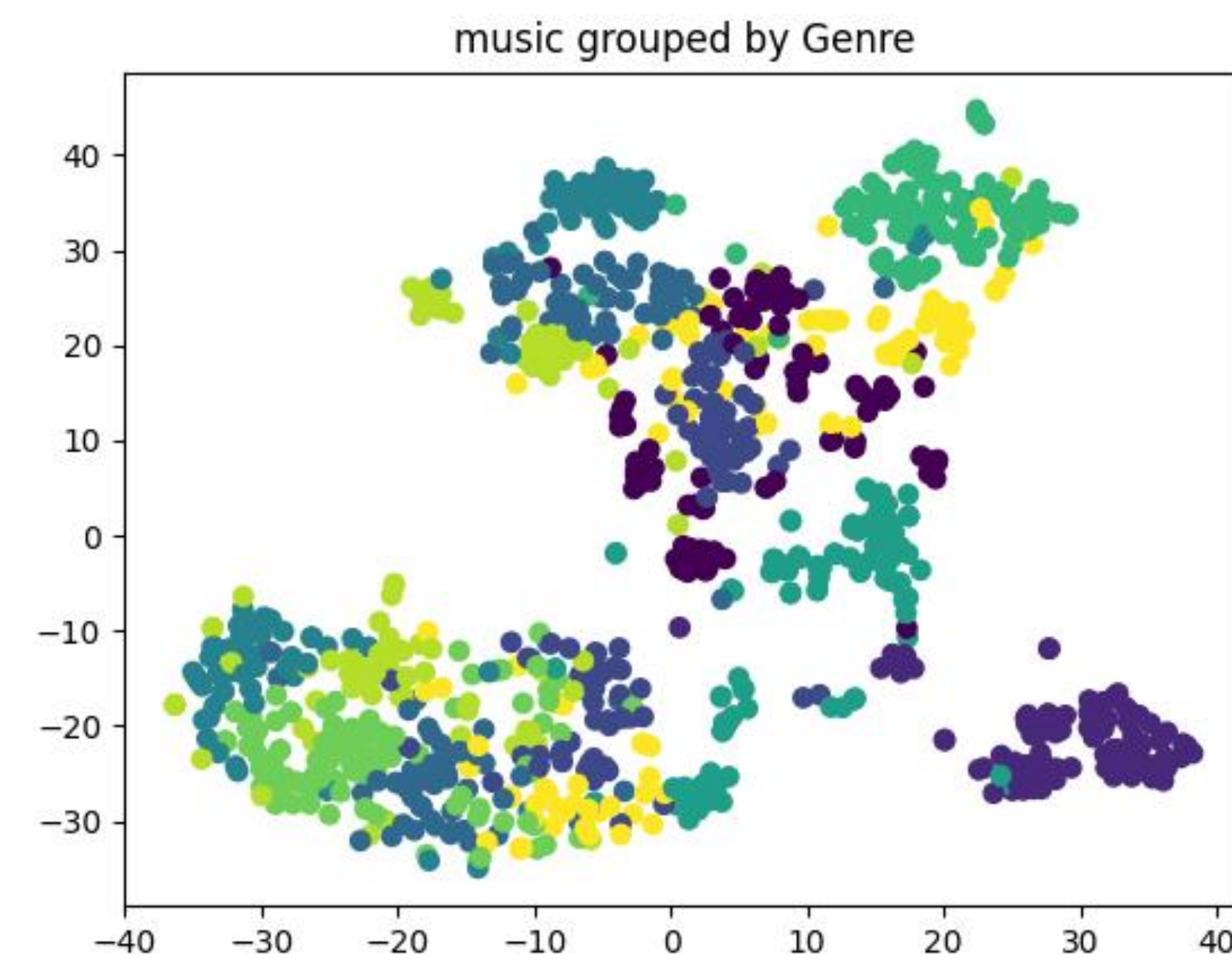
The genres included in the dataset are blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, and rock

Challenges and Limitations

- Large open/free datasets of popular and modern music are hard to find
- Embedding songs as vectors using a transformer can be computationally expensive
- Performance depends heavily on how well the transformer is trained
- Not many quantitative performance metrics
- Genres are often subjective and may not definitively capture the full spectrum of musical styles.

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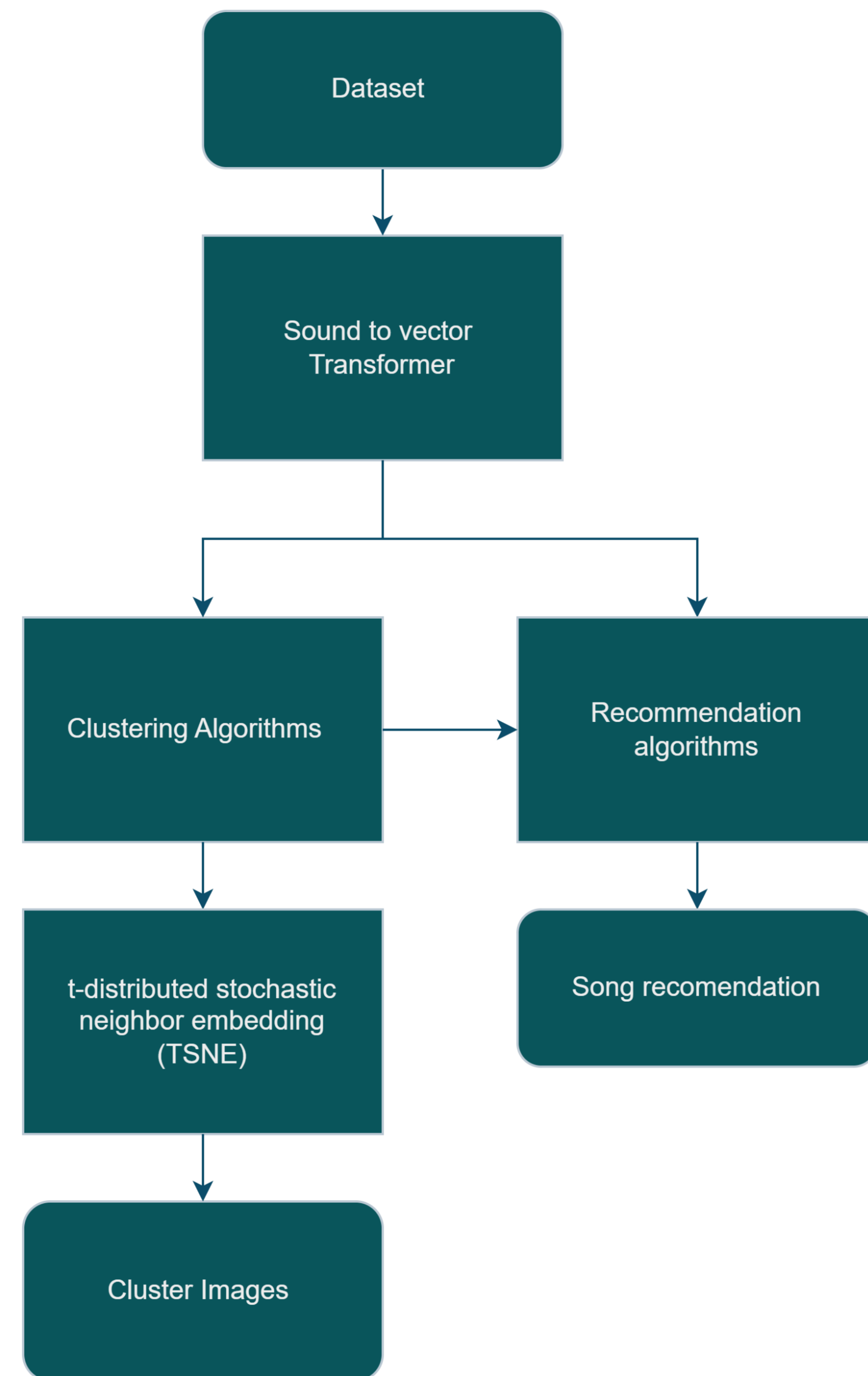


Transformers

Transformers are a semi-supervised machine-learning architecture that transforms an input sequence into an output sequence. Some of the key characteristics of a transformer are the encoder-decoder architecture and attention mechanisms. Transformers are commonly used for Natural language processing (the T in GPT stands for Transformer).

t-SNE

t-SNE is a non-linear dimensionality reduction technique primarily used for visualizing high-dimensional data in a low-dimensional space (typically 2 or 3 dimensions). It's particularly effective at preserving local structure in the data



Clustering Algorithms

Clustering is a Common machine-learning task that seeks to group data into categories. Some of the Clustering algorithms tested in this research project include K-means, affinity propagation, Agglomerative, DBSCAN, and HDBSCAN.

Recommendation Algorithms

the purpose of the recommendation algorithms is to take in a subset of liked songs and then return to the user a recommendation for a song that is similar to the liked songs. Things like this could assist in playlist generation.

Min Sum of Distance

this algorithm works by calculating the Euclidian distance from the liked song to all of other songs then adding the distances to each point then recommending the one with the smallest sum of Distances.

First Song in Common

this algorithm works by creating rankings (based on Euclidian distance) for each of the liked songs to every other song returning the first song all of the rankings have in common

Max Sum of Cosine Similarity

this algorithm works by calculating the sum of cosine similarity between each liked song and all other songs. then it recommends the song with the most similarity

Conclusions and Further Research

this research project was successful in the following areas:

- embedding variable-length music clips as 512-dimensional vectors
- Visualizing music embedding in 2-d plain
- Grouping songs with similar features with an array of clustering algorithms
- Recommendation algorithms were successfully implemented

Some of the things that could be explored in a follow-up research project include:

- Fine tuning the Transformer model for the dataset
- Using a larger dataset with modern music
- Experimenting with different size vector embedding
- Implementing recommendations with real playlists
- Implement better metrics to assess performance